University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab

LAB REPORT

Experiment Number: 03

Experiment Title: Distance Measurement using Ultrasonic Sensor and Servo Motor

Submitted by:

Name : Sunandan Das

Student ID : 23101218

Section : D2

Submitted to:

Zaima Sartaj Taheri
Lecturer,
Department of Computer Science and Engineering

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1. Experiment Name

Distance Measurement using Ultrasonic Sensor and Servo Motor.

2. Objective

To design and implement a distance measurement system using an ultrasonic sensor and a servo motor, with visual feedback provided through LEDs. The system detects the proximity of objects within a specified range and responds by adjusting the servo position and illuminating corresponding LEDs. 3.

3. Hardware & Software Requirements

Hardware Requirements:

- Arduino Uno R3
- Ultrasonic Distance Sensor
- 3 LEDs (Red, Yellow, Green)
- Servo Motor
- Breadboard
- Jumper Wires
- 3 Resistors
- USB Cable with power supply

Software Requirements:

- Arduino IDE (for coding and uploading the program)
- Circuit design software (Tinkercad) for simulation and schematic diagrams.

4. Circuit Diagram / Schematic

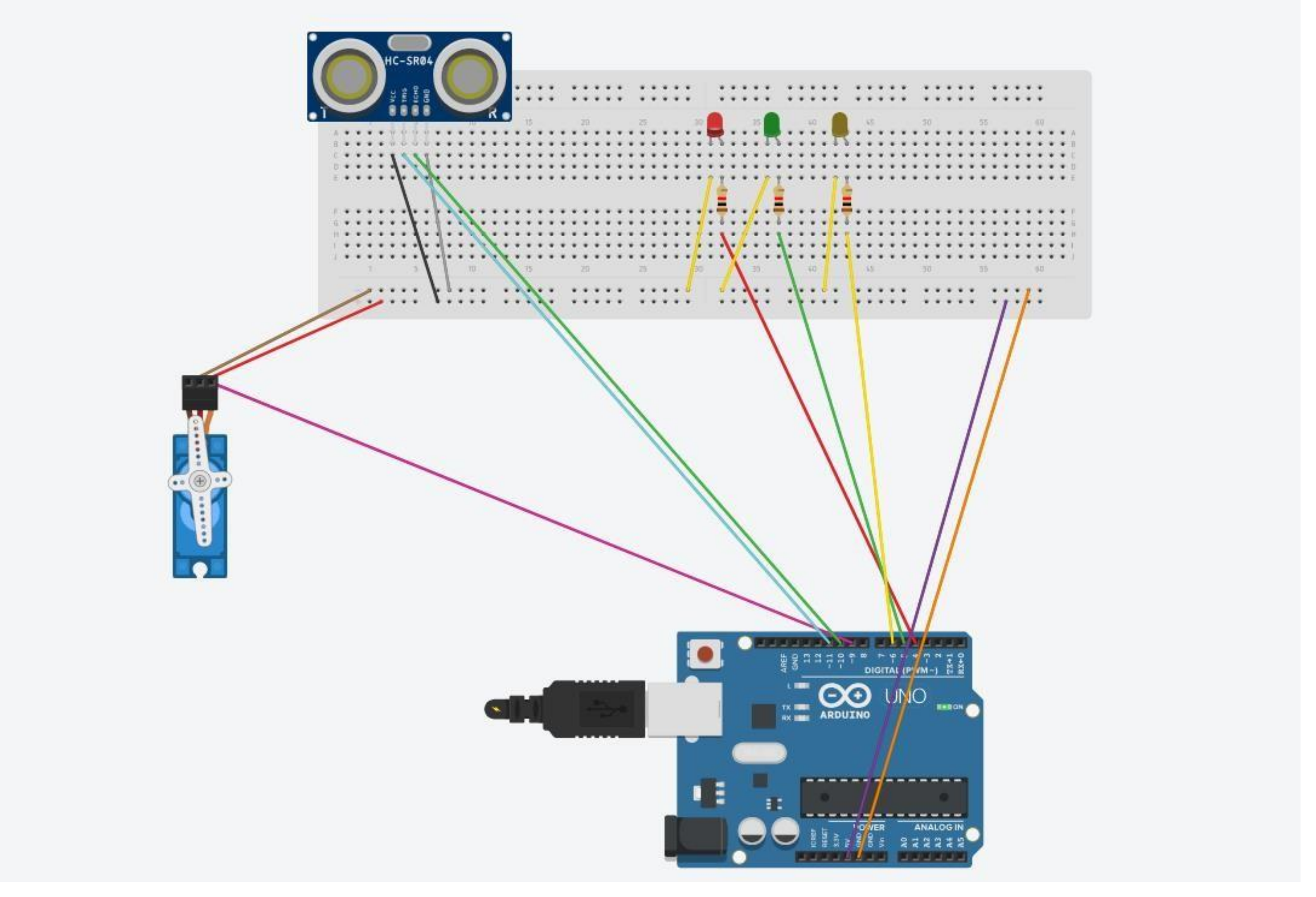

Figure-01: Distance Measurement using Ultrasonic Sensor and Servo Motor

5. Code / Assembly Program

```
#include <Servo.h>

Servo myServo;


// Pin assignments const

int trigPin = 11; const int

echoPin = 10; const int

greenLED = 5; const int

yellowLED = 6; const int
```

```
redLED = 4; long
duration; int distance;

void setup() {
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);

  pinMode(greenLED, OUTPUT);
pinMode(yellowLED, OUTPUT);   pinMode(redLED,
OUTPUT);

  myServo.attach(9);
  Serial.begin(9600);
}

void loop() {   // Send
ultrasonic pulse
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);

  // Measure echo time   duration =
pulseIn(echoPin, HIGH);

  // Calculate distance in cm
distance = duration * 0.034 / 2;
Serial.print("Distance: ");
```

```
  Serial.print(distance);
  Serial.println(" cm");

  // Control LEDs and Servo based on distance
if (distance <= 5) {
digitalWrite(greenLED, LOW);
digitalWrite(yellowLED, LOW);
digitalWrite(redLED, HIGH);
myServo.write(90);
  }    else if (distance <= 10) {
digitalWrite(greenLED, LOW);
digitalWrite(yellowLED, HIGH);
digitalWrite(redLED, LOW);
myServo.write(45);
  }

  else {
    digitalWrite(greenLED, HIGH);
digitalWrite(yellowLED, LOW);
digitalWrite(redLED, LOW);
  myServo.write(0);
  }

  delay(100);
}
```

6. Output / Observations

Upon execution, the system consistently measured object distances and displayed the readings on the Serial Monitor. The behavior observed was as follows:

a) **Distance ≤ 5 cm:** The red LED illuminated, and the servo rotated to 90°, indicating a close detection.

b) **5 cm < Distance ≤ 10 cm:** The yellow LED illuminated, and the servo rotated to 45°, signaling a moderate range.

c) **Distance > 10 cm:** The green LED illuminated, and the servo rotated back to 0°, marking a far-range detection.

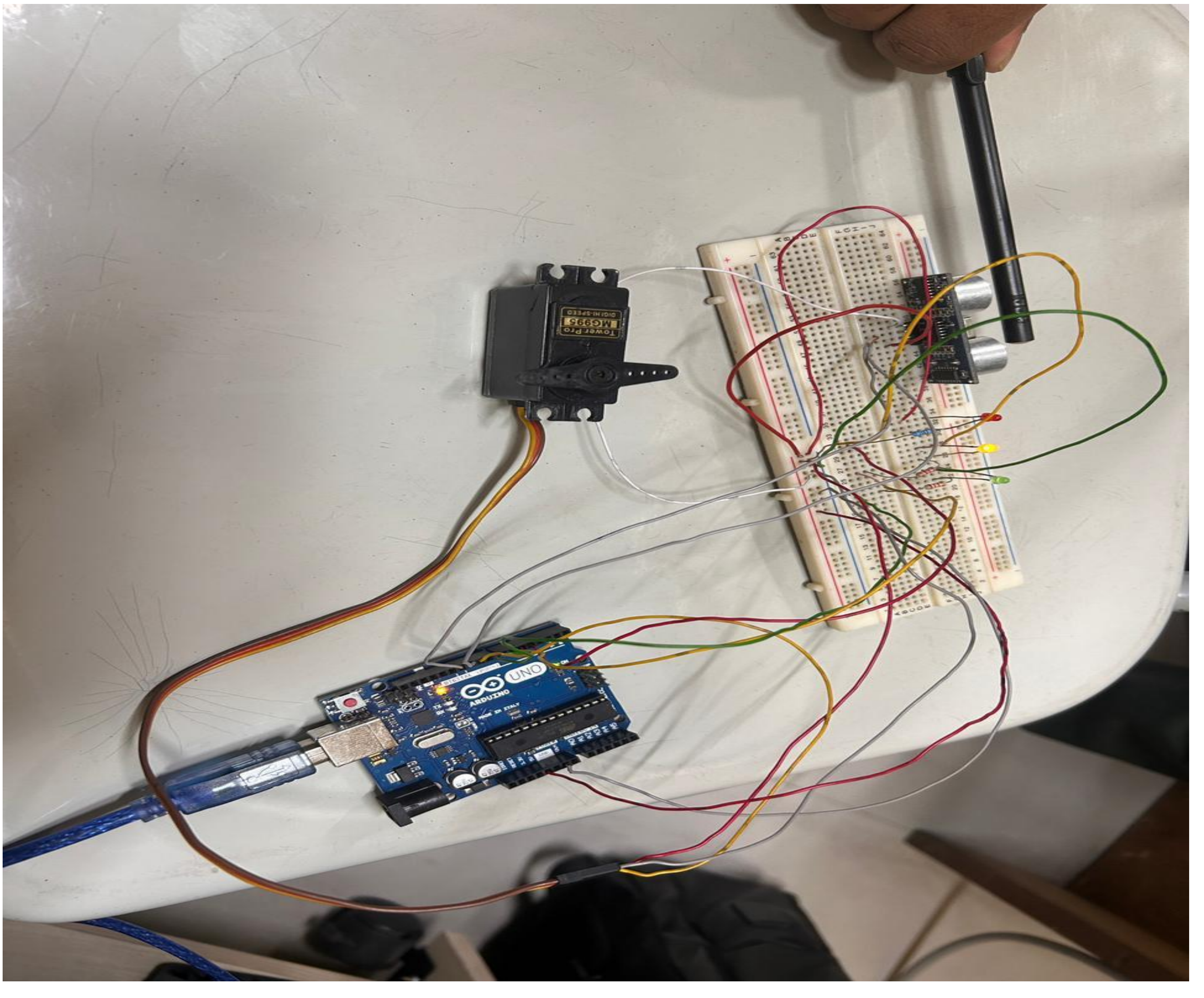

Figure-02: 6-10 cm, Yellow LED ON

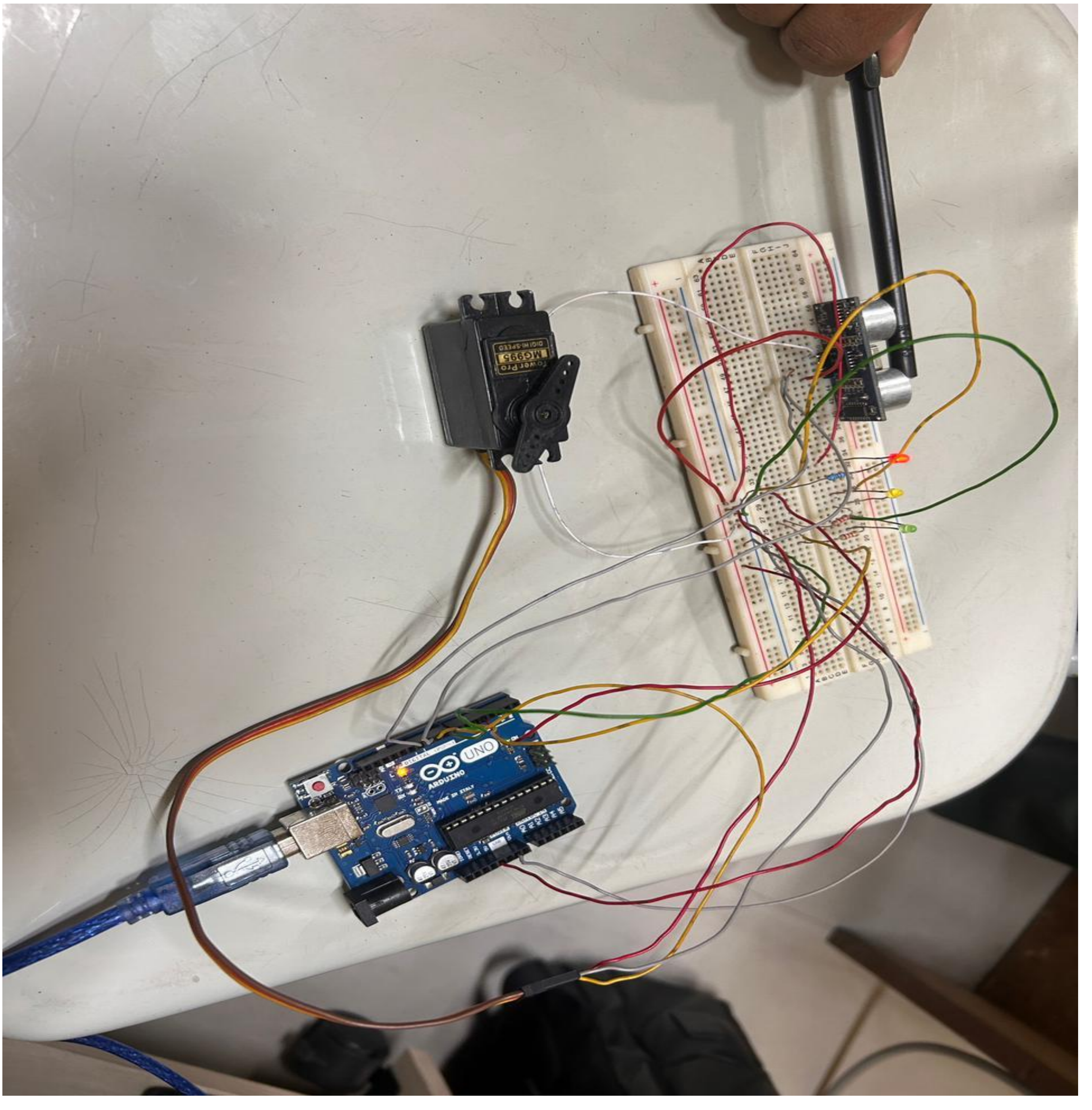

Figure-03: ≤ 5 cm, red LED ON

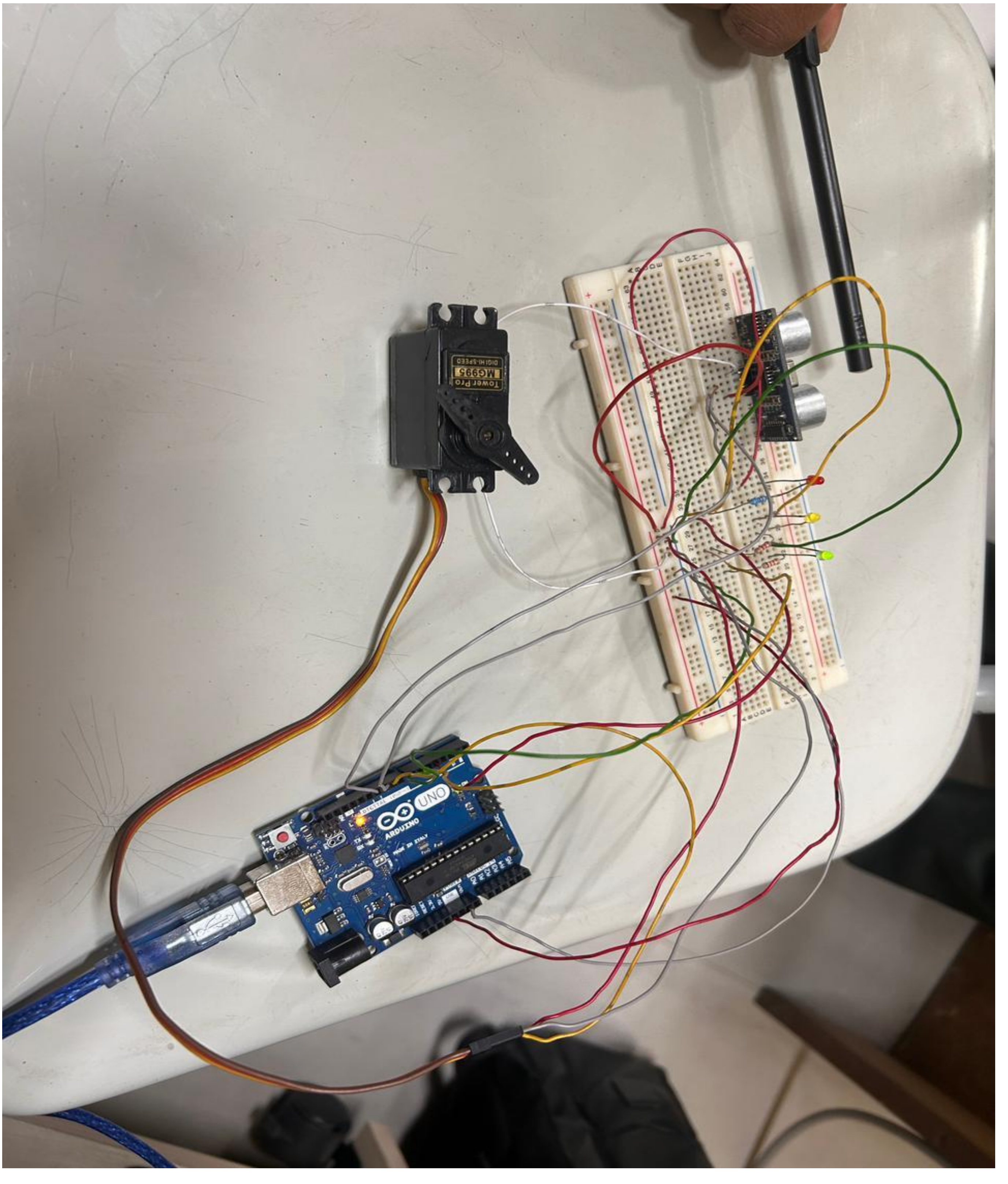

Figure-04 : > 10 cm, Green LED ON

The responses were stable, with transitions occurring promptly as the detected distance changed. This confirmed that the ultrasonic sensor and servo were functioning in sync with the programmed logic.

7. Result:

The project implementation successfully demonstrated a functional distance-based detection and response system using an ultrasonic sensor, LEDs, and a servo motor. The system was able to measure the distance of nearby objects and respond according to the predefined distance ranges programmed in the microcontroller.

During testing, the system behaved as expected. When an object was at a distance of 5 cm or less, the red LED turned on and the servo motor rotated to 90°, indicating that the object was very close. When the measured distance was between 6 cm and 10 cm, the yellow LED illuminated and the servo moved to 45°, representing a medium distance. For distances greater than 10 cm, the green LED was activated and the servo returned to 0°, signaling that the object was far away.

The Serial Monitor readings verified that the ultrasonic sensor measured the distance accurately, and the LEDs responded correctly according to the measured values. Initially, a technical issue occurred because the servo motor did not operate properly during the first test. After troubleshooting, the faulty servo motor was replaced with a new servo, and the system then functioned correctly with smooth and synchronized responses.

Overall, the experiment successfully met its objective of developing a microcontroller-based distance measurement system that provides both visual feedback through LEDs and mechanical response through a servo motor.

8. Conclusion:

This experiment demonstrated how a microcontroller can integrate sensing, processing, and actuation to create a complete functional system. The ultrasonic sensor continuously measured the distance of nearby objects, and based on these measurements, the system produced clear feedback through LEDs and the movement of a servo motor. While performing the experiment, a problem occurred when the servo motor initially failed to operate. After identifying the issue, the faulty servo was replaced with a new one, and the system then functioned properly with synchronized LED indications and servo movement. The Serial Monitor readings also confirmed that the distance measurements were accurate. Through this project, I gained practical experience in applying conditional logic, sensor integration, and timing control in embedded systems. Although the setup was simple, the same principles are widely used in real-life applications such as parking assistance systems, automatic doors, and basic robotic systems. Overall, the project successfully achieved its objective and helped strengthen my understanding of combining hardware components with programming to solve practical problems in automation and control.